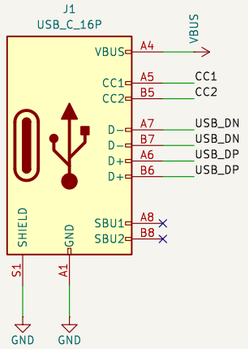
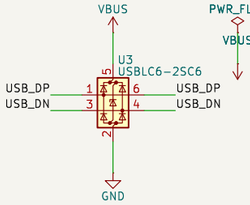
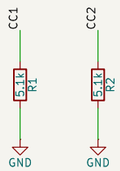
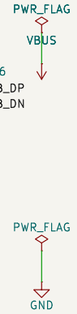
USB-C input, CC pull-downs, D+/D- ESD clamp

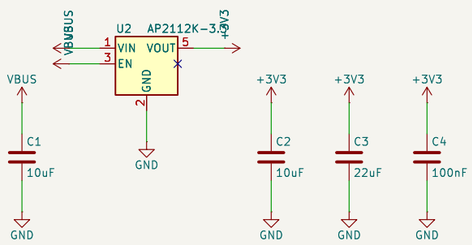




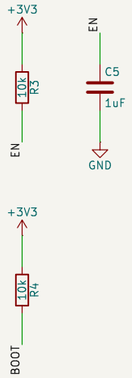




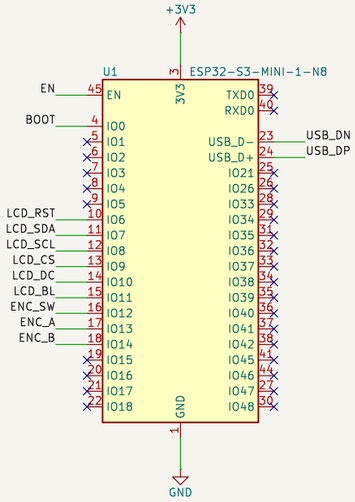
5V -> 3.3V LDO (AP2112K, 600 mA)



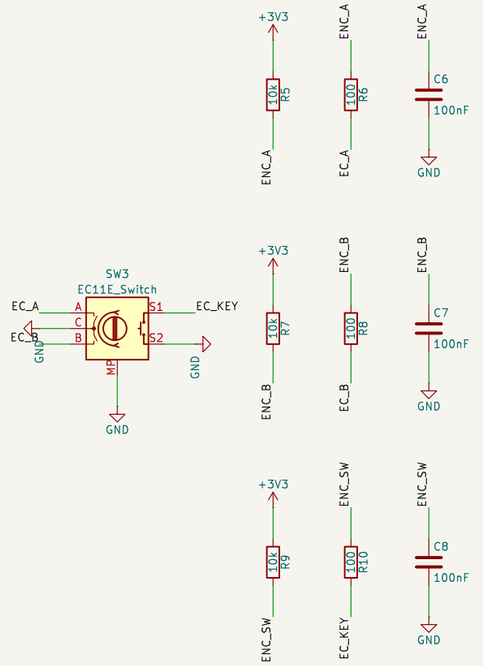
EN power-on reset RC and BOOT key



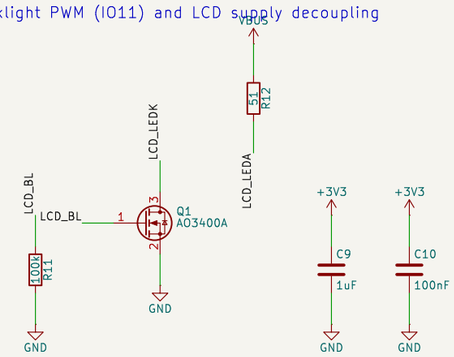
ESP32-S3-MINI-1-N8 (no PSRAM)
Strapping pins IO3/IO45/IO46 left unconnected



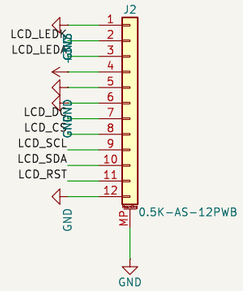
EC11 encoder, RC debounce ~1 ms



LCD backlight PWM (IO11) and LCD supply decoupling



LCD FPC: T128HC-C12-05 (GoldenMorning), 12P 0.5mm, FPC 0.3mm
J2 top contact: the FPC folds 180 deg under the panel,
contacts face up; J2 pin n = FPC pin n
R12: (5.0V - 3.0V) / 51R = 39 mA (panel IF = 40 mA typ)



ESP32-S3-MINI-1-N8 / EC11 / GC9A01 round LCD (FPC)

Sheet: /
File: knob-v2.kicad_sch

Title: Knob Board V2

Size: A3 | Date: | **Rev: 2.0**
KiCad E.D.A. 9.0.8 | Id: 1/1